

Engr. Jalal Khan

NEOM Approved Electrical Engineer | Site Installation | Testing & Commissioning |

Mecca city, KSA | [Email Address Jalalkhan10087@gmail.com](mailto:Jalalkhan10087@gmail.com) | [Contact: 050 631 8942](tel:0506318942) | [Linkedln](#) |

Profile Summary:

NEOM Approved Electrical Engineer with 6+ years of experience in LV & MV (13.8 to 33 KVA) systems. Skilled in handling installation, testing & commissioning, troubleshooting, fault finding, maintenance of various types of transformers, switchgear, AIS/GIS/VCB Circuit breakers, MDBs, cable testing & underground fault locator, Protection relay fetching and setting (RET615 & REF615), along with thermal relays and other types of protection testing and troubleshooting, and conducting duct banks. Hands-on experience in panel termination and installation. Strong in scheme matching, accuracy testing, measuring perimeters, SLD review, IFC/as-built drawings, BOQs, load flow & fault analysis. Proficient in AutoCAD Electrical, Revit MEP, MATLAB & PSpice & other testing-related software (such as Omicron, Advance CT analyzer & many more). Holds PEC/HEC Registered | Washington Accord Level II | Assessed by Engineers Australia.

Core Skills & Abilities:

- Transformer, RMU & MV/LV Switchgear Testing & Commissioning
- GIS, ACB, VCB Breakers Testing & Maintenance
- MDBs, MCCs, ATS Panels – Installation & Testing
- CT/VT Testing, Relay Testing & Protection Coordination
- Cable Testing, Meggering, Hi-Pot, Cable Route & Fault Location
- VFDs, Soft Starters & Motor Testing
- Earthing, Arc Protection & Surge Protection Systems
- Panel Termination, Scheme Matching & Fault Diagnosis
- SLD, BOQ, IFC/As-Built Review & QA/QC Reporting

Testing Instruments Proficiency:

- Omicron CPC-100, TestRano 600, CIBANO 500, CT Analyzer
- FREJA 300, SEVERKER 900/780
- ABB ETRIP Kit, FRAX-150, SACE PR010/T
- Megger (up to 5 kV), Micro Ohmmeter (MOM)
- AC Hi-Pot Tester, HV Test Kit, Vacuum Tester (VIDAR)
- SF6 Gas Analyzer

Software & Technical tools:

- AutoCAD Electrical, Revit MEP
- MATLAB / Simulink, PSpice, ETAP (Basic), MATLAB
- ABB PCM600 (RET615 / REF615)
- Omicron Test Universe, Advanced CT, ABB Ekip
- Megger PowerDB
- MS Office Suite

Professional Work Experience (With Project Roles & Responsibilities):

Electrical Engineer - Saudi Arabian Octavo Company Limited

Project: Third Saudi Expansion of the Holy Mosque & Surrounding Areas

Client: Dar Engineering | Contractor: SBG | Dates: [June 2025 – till today]

- Executed testing and commissioning of MV & LV electrical systems, including transformers, RMUs, Switchgears, MCC & ATS & PFC panels, and cable testing as per SEC standards.
- Performed technical review and verification of scheme, single line diagrams (SLDs), and coordination with design teams for accurate implementation of testing procedures after installation on-site.

- Performed testing for 7.5hp & 15hp motors for both soft & VFDs using designated devices, communicating with consultants for proper complications, and by following the project standards
- For LV side testing: UDPs, EDPs, UPS, SAS, with dry-type transformers, LCP panels, and DC control related circuits for proper control by BMS & SCADA at time of use.

Electrical Engineer – Al-Shalawi International Co. for Trading & Contracting

Dates: [October 2023 – May 2025]

- Supervised installation and testing of MV cable networks, including laying, jointing, termination, and insulation testing up to 33kV.
- Executed Transformer & incoming feeders' installation for system 132KV to supply or feed 66kV system based on GIS.
- Managed installation and commissioning of substation equipment – RMUs, transformers, MDBs, grounding systems, and control panels.
- Handled electrification of irrigation systems, pump/control panel installations, and integrated automation with power supply infrastructure.
- Reviewed SLDs, IFC/shop drawings, and ensured compliance with NEOM/SEC/IEC standards during installation, testing & commissioning.

Electrical Engineer – SHARHAN ENERGY CO. LTD

Dates: [August 2021 – October 2023]

- Maintained and submitted project documentation, including daily reports, test reports, punch lists, as-built drawings, and commissioning records for client approval.
- Executed HV power cable installation, including HDPE ducting, cable pulling, jointing, glanding, and terminations.
- Installed and commissioned motor control centers (MCCs), star-delta starters, VFDs, and irrigation pump control panels with interlock wiring.
- Conducted cable insulation resistance testing, earth continuity verification, loop impedance, and merger testing for system integrity.

Electrical Project Engineer – Greenwendenergy Private Limited

Dates: [June 2020 – July 2021]

- Supervised the installation of LV systems.
- Ensured accurate installation of structured cabling systems for panels, distribution panels, and Auxiliary supply.
- Verified the installation of power distribution and control systems in coordination with SLD layouts, ensuring compatibility with automated and smart building technologies.

Completed projects:

- Al-Shamiyah Haram Extension & Mataf Project Substation, Makkah, KSA
- 110KV - 33KV Substation Project, KSA - SEC Consultants.
- MERSEKS Logistic Park Substation SS5, Jeddah, KSA, SIE Consultants.
- Work Order 26 – Construction of Signalized U-Turns on Highway 55, Neom, KSA
- Construction of HW392 Trumpet Interchange 1,2 & 3, Package – 3, Neom KSA
- 132KV GIS mobile substation Installation, Neom KSA
- Plant equipment and backup cluster, Neom KSA
- PESCO maintenance Project of 132KV to 33KV grid station, Peshawar, PK
- Solar Project: Installation of a complete 1.5 MW solar system with testing and commissioning for the Data Center unit supply, Peshawar, PK. (FWO Client)

Other projects:

- Three-phase star-connected load, symmetric ohmic load
- Designing a step-down transformer with greater efficiency

- Modification of a non-hybrid inverter to maximize the efficiency of an off-grid system
- Smart home automation, Integrated with a solar system for efficient use of energy.
- Modification of a non-hybrid inverter to maximize the efficiency of an off-grid system.
- Designed the Circuit and PCB Layout for the Smart Energy Meter
- Single To Dual Output Smart Load Device (PCB Design).
- Optimizing Solar-integrated Smart Home Automation for Energy Efficiency & Sustainability.

Licenses & certification:

- *Pakistan Engineering Council Code: ELECT/103773 -PEC*
- *Engineers Australia - Professional Engineer Code: 520758*
- *ISO 9001:2015 (Quality Management System)*
- *Driving Licence*

Qualification:

“Bachelor of Science in Electrical Engineering”

University of Engineering and Technology, Peshawar, Pakistan.

“Diploma of associate in electrical”

Technical & Education Council, Islamabad, Pakistan.

Recognized by: Engineers Australia (EA) “Professional Engineer” | Level 2 Washington Accord Recognition
|Pakistan Engineering Council (PEC) | Higher Education Commission (HEC) |

Achievements:

- “Certificate of Excellence” (At Project WO-26) presented by NEOM’s PM.
- “Certificate of Excellence” (At Project IC-1, 2 & 3) presented by Consultant Bechtel’s PD.
- Awarded “Best Speaker” at the International ISE EXPO.
- “Best Project Poster Competition” Winner Certification at the International ISE EXPO.
- Secured highly competitive Final Year Project Funding from the “Government funding IGNITE” Under Project Code: NGIRI 2023-22301
- Displayed exemplary team leadership and coordination skills during FYP, contributing significantly to project success.
- “Certificate of appreciation” for the best Project Ideas at UET Peshawar.
- “Certificate of participation” for playing a crucial role in the Solar Seminars debate at UET Peshawar.

Courses:

- Introduction to Transmission Control Protocol/Internet Protocol TCP/IP
- Electric Power Systems -University at Buffalo
- Internet of Things (IoT) devices
- Embedded software and hardware architecture
- Converter Circuit -The University of Colorado Boulder
- MV Switchgear Schematics Course: Tripping, Trip Circuit Supervision, Interlocking, and Indication Circuits

References:

(References are available upon request)